

## Problem

A transformer having 2,400 turns on the primary and 48 turns on the secondary is used as an impedance-matching device. What is the reflected value of a 3- $\Omega$  load connected to the secondary?

## Step-by-step solution

## Step 1 of 1

$$\begin{aligned}
 n &= \frac{N_2}{N_1} \\
 &= \frac{48}{2400} = \frac{1}{50} \\
 Z_{TH} &= \frac{Z_L}{n^2} = \frac{3}{\left(\frac{1}{50}\right)^2} = 7.5 \text{ K}\Omega .
 \end{aligned}$$